

Salinity stress on agar

Overview

Applying various salinity stress levels to Arabidopsis on agar to evaluate which concentration of trehalose protectant provides the best rescue effectiveness.

A. Materials

a. Agar

- i. $\frac{1}{2}$ MS medium powder
- ii. MES buffer
- iii. Distilled water
- iv. Sucrose
- v. Agar powder
- vi. pH meter
- vii. 2000ml beaker
- viii. Hot plate
- ix. Magnetic stirring rod
- x. Aluminum foil

b. Transfer

- i. Tweezer
- ii. New sets of neutral agar

c. Apply treatments

- i. Trehalose
- ii. NaCl solution
- iii. Sterile ddH₂O
- iv. Incubator
- v. Rack

B. Procedure

A. $\frac{1}{2}$ MS medium agar for vernalization

1. The following protocol is for 750ml of agar
 - 15 neutral agar plates
2. Prepare the following:

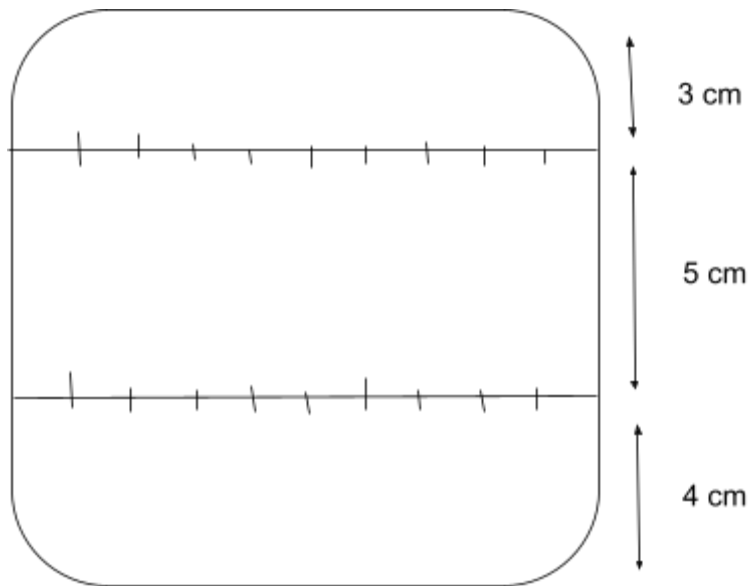
Ingredient	
$\frac{1}{2}$ MS medium power	1.725g
MES buffer (0.5g/L)	0.375g
Distilled water	750ml

Sucrose (1%)	7.5g
Agar powder (1%)	7.5g

3. Add the contents into a 2000ml beaker
4. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
5. Use the pH probe and measure the pH of the solution, make sure the pH is between 5.7-5.8 (Use KOH to adjust the medium pH)
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top
7. Autoclave for 20 minutes at 127°C
 - a. Wait for the pressure to reach 0 before retrieving the flask
8. During the autoclave, label all of the petri dishes with a marker in the biosafety cabinet
 - a. right top: date and arrow pointing up to show direction for pictures
9. Transfer half of the flasks to BSC
 - a. Store the other half of the flasks in the water bath while pouring so that it does not solidify
10. Make sure its surface is dry by wiping the flasks with a paper towel before spraying 70% EtOH on them
11. Pour each of the medium mixtures into their square petri dishes (around 50ml)
12. Transfer the other half of the flasks into the BSC
13. Pour each of the medium mixtures into their square petri dishes
14. Let the petri dishes solidify at room temperature under the BSC

B. Planting and treating arabidopsis seeds

1. Open the BSC
2. Use the seed template (3cm, 5cm, 4cm) to plant 18 seeds on each agar plate in two rows, nine seeds on each row



3. Use breathable tape to seal each of the agar plates
4. Label the vernalization date
5. Wrap the agar plates with aluminum foil
6. Vernalize in the 4°C refrigerator for 4 days
7. Incubate in the growth chamber for 4 days

C. 1/2 MS agar plate preparation(salinity test trial plates)

The following protocol is for 600ml of agar

- 12 agar plates
 - control(no trehalose) x control(no NaCl)
 - control(no trehalose) x 150mM NaCl
 - control(no trehalose) x 250mM NaCl
 - 1mM trehalose x control(no NaCl)
 - 1mM trehalose x 150mM NaCl
 - 1mM trehalose x 250mM NaCl
 - 5mM trehalose x control(no NaCl)
 - 5mM trehalose x 150mM NaCl
 - 5mM trehalose x 250mM NaCl
 - 10mM trehalose x control(no NaCl)
 - 10mM trehalose x 150mM NaCl
 - 10mM trehalose x 250mM NaCl

1. Prepare the following:

½ MS medium powder(w/o sucrose)	1.38g
MES buffer	0.3g

Distilled water	600ml
Sucrose(1%)	6g
Agar powder(1%)	6g

2. Add ½ MS medium powder (w/o sucrose), MES buffer, distilled water, and sucrose into a 2000ml beaker.
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8
 - a. If the pH is too low, add 0.1M KOH slowly to increase it
5. Pour 200ml of mixture out into another flask to separate into three flasks, label with [0mM], [150mM] and [250mM]
6. Add 2.192 g of NaCl into the flask labelled [150mM NaCl]
7. Add 3.652 g of NaCl into the flask labelled [250mM NaCl]
8. Prepare 4 flasks for different trehalose concentrations(0, 1, 5, 10mM)
9. Prepare 12 flasks and pour the respective trehalose and NaCl concentration in each of the flasks
 - a. making sure there is 50ml in each of the flasks(12)
 - b. label each flask clearly according to their trehalose concentrations
10. To their respective trehalose concentrations, add

Trehalose concentration	Amount of trehalose (for 50ml of medium)
0mM	0µL
1mM	500µL
5mM	2500µL
10mM	5000µL

11. Add 0.5g of agar into each flask.
12. Wrap the necks of the flasks with aluminum foil and stick a piece of indicator tape on top
13. Autoclave for 20 minutes at 127°C
 - a. Wait for the pressure to reach 0 before retrieving the flask
14. During the autoclave, label 8 of the petri dishes with a marker in the biosafety cabinet
 - a. right top: date and arrow pointing up to show direction for pictures
 - b. bottom lid: different trehalose concentrations and NaCl concentration or control
15. Transfer half of the flasks to BSC
 - a. Store the other half of the flasks in the hot bath while pouring so that it does not solidify
16. Make sure its surface is dry by wiping the flasks with a paper towel before spraying 70%

EtOH on them

17. Pour each of the medium mixtures into their square petri dishes (around 50ml)
18. Transfer the other half of the flasks into the BSC
19. Pour each of the medium mixtures into their square petri dishes
20. Let the petri dishes solidify at room temperature under the BSC

D. Transferring arabidopsis

1. Take the stack of agar plates wrapped in aluminum foil out of the refrigerator
2. Open the BSC
3. Spray alcohol on the agar plates before placing them inside the BSC
4. Use a marker to draw a straight line along a ruler 3cm from the top of the plate using the seed template
5. Draw 8 dots at the ends of the roots
6. Place the square plates vertically at a slight angle
7. Choose 8 germinated seeds after vernalization that have similar root lengths or germination status and align the endpoints of their roots to the line drawn on the plate
8. Seal the plates with breathable tape and label the incubation date with a marker before placing the vertical plates on the stacks in the growth chamber
 - a. Slightly tilted towards the back

E. Recording(every day)

1. Observe for 10 days
2. Set up the black background platform and turn on the lights
3. Take out the agar plates one by one out of the growth chamber and place them on the black background platform(bottom lid facing up)
4. Take pictures(clearly record the labels, no reflections and avoid water vapor from covering the view of the plants)
5. Place the agar plates back, tilted slightly to the back on the rack, in the growth chamber
6. weigh the fresh weight

Salinity Stress Test

